What’s New with .NET MAUI

.NET Multi-platform App UI (MAUI)

Single project system, single codebase



WinUI



Mac Catalyst

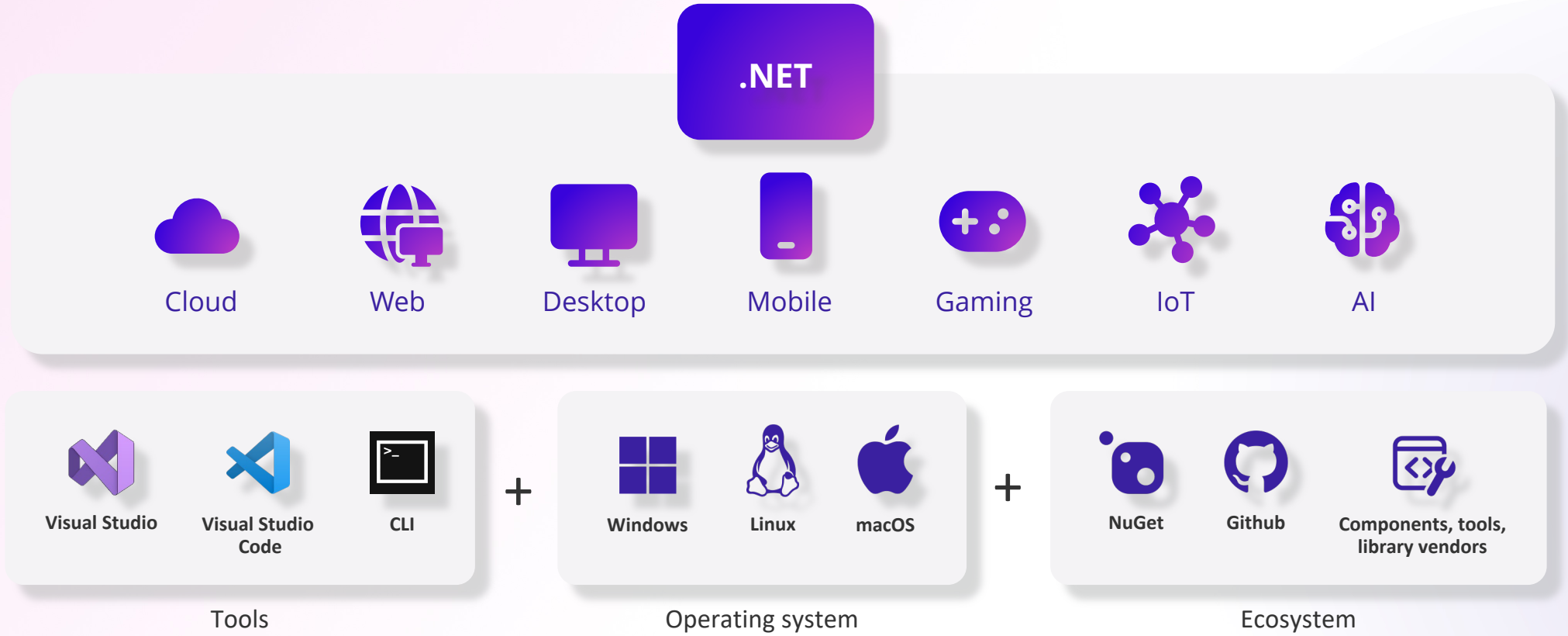


iOS



Android

Build anything with a unified platform



Hello, .NET 8 MAUI!

Top 2

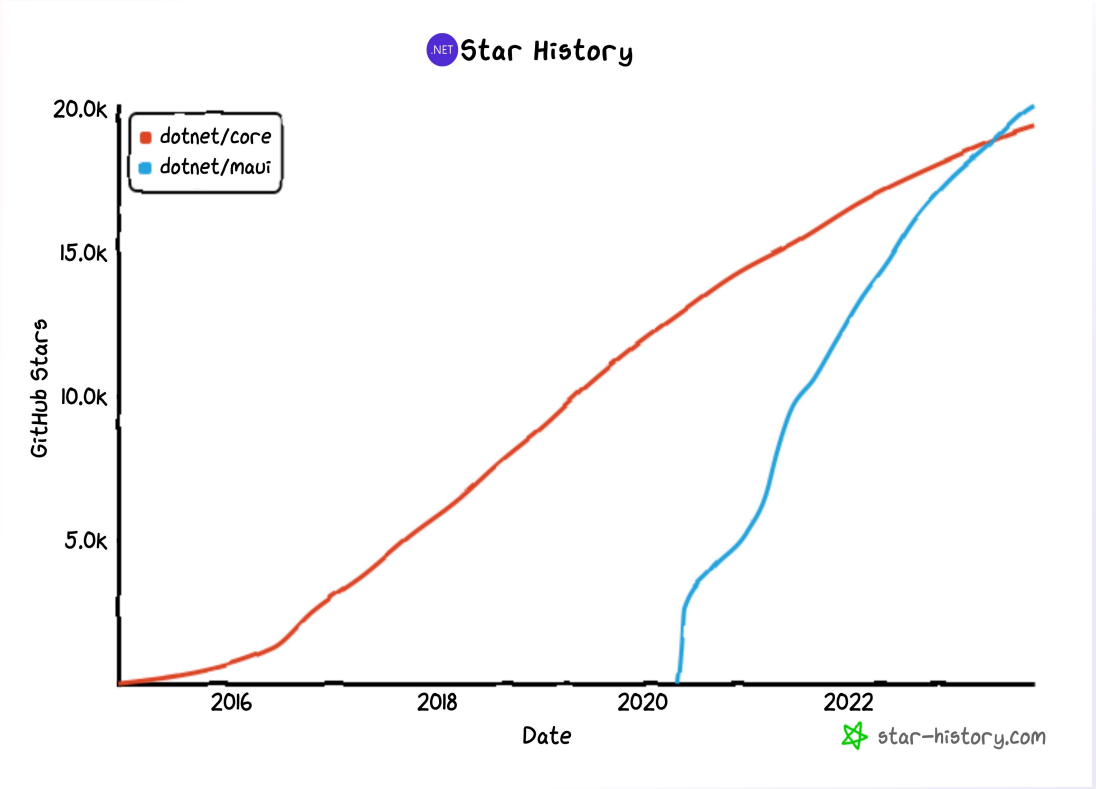
Active GitHub users in dotnet

Top 2

Most active in .NET Foundation

Top 5

Most active C# repo on GitHub October 2023



1600+ PRs and **600+ Issues Closed** in .NET 8

github.com/dotnet/maui


Retail Insight

3M

hogeschool
vives

irth
UtiliSphere

S


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OEXA

MAHIZ
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Microsoft
Dynamics 365

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CIVICA

SPORTS
NEXT
SPORTS x TECH

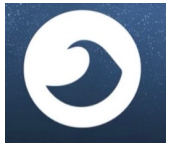
DevInMotion

Belgian
Red Cross
Flanders

ambientindustries

A

insight

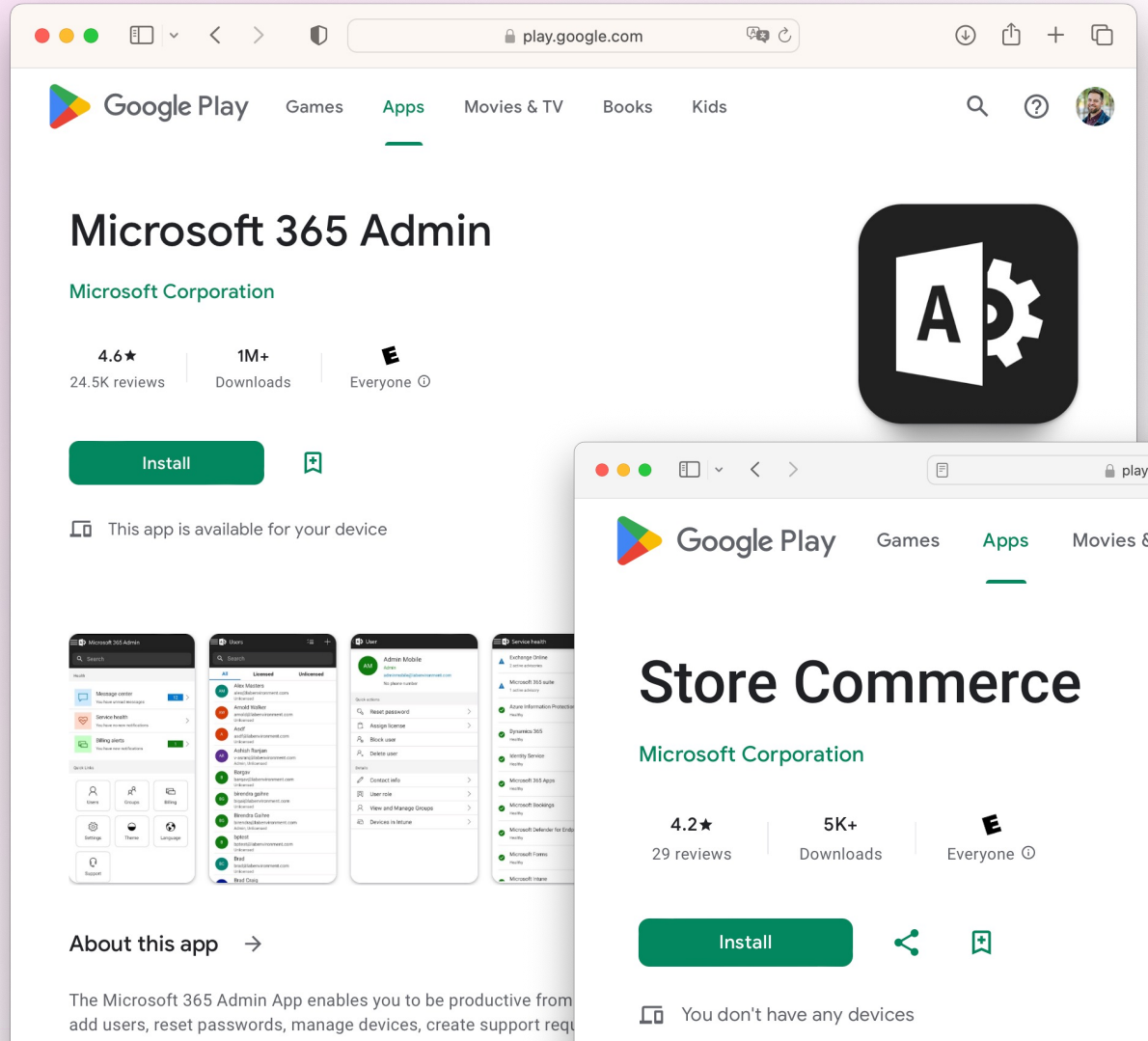
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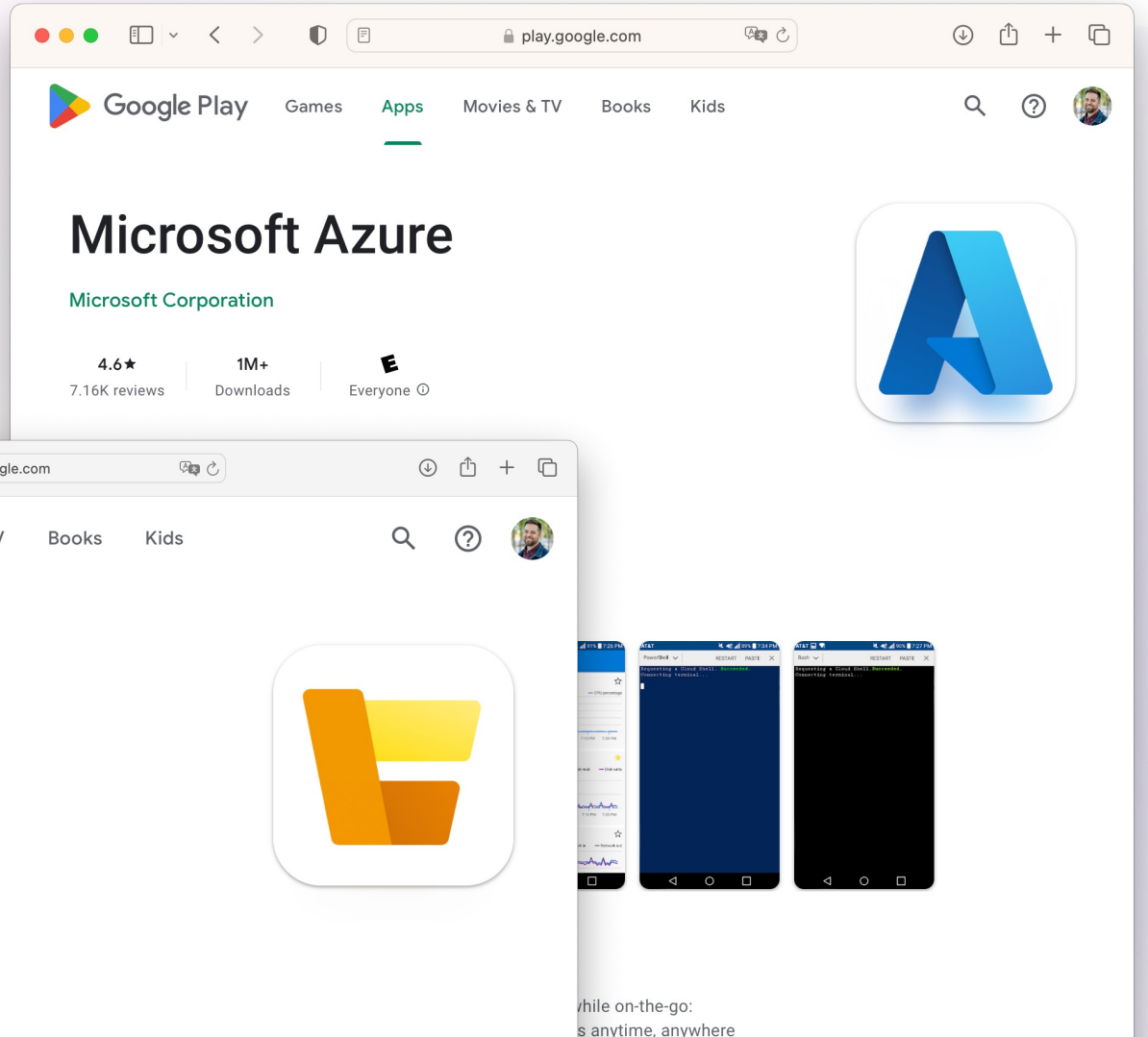
Soluzione 1

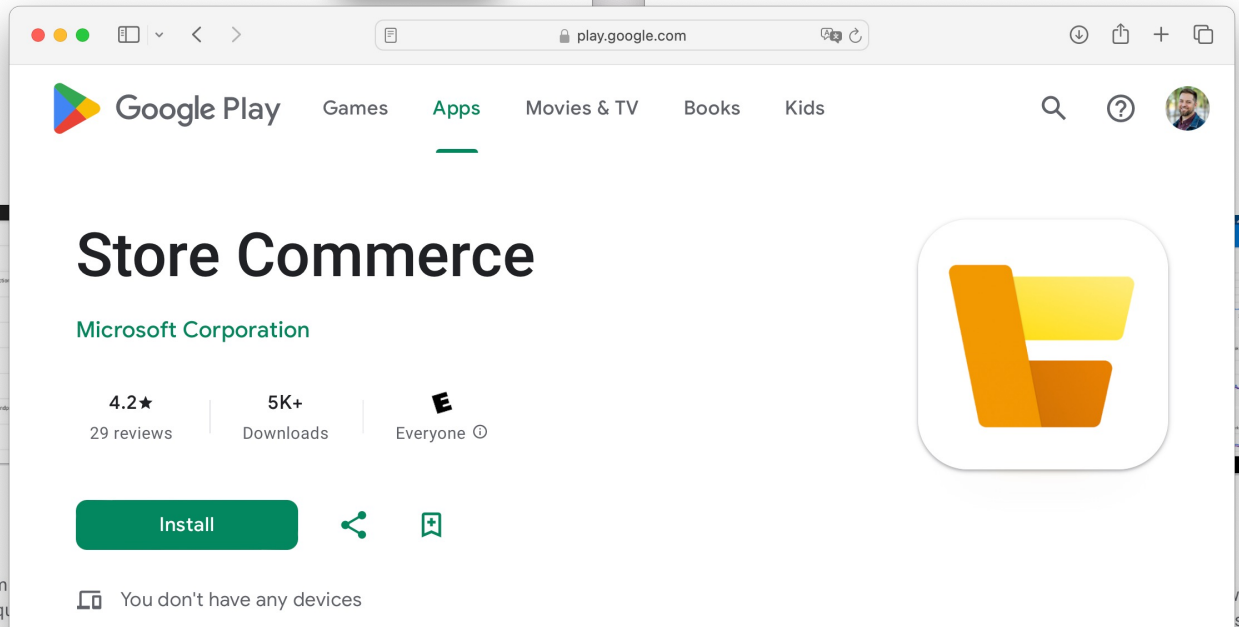
REY
TECHNOLOGY

[A]


.NET MAUI at Microsoft







RELEASED

.NET MAUI in .NET 8

- Improved performance and memory management
- New hybrid and desktop app features
- Flexible tooling for development
- Tons of bug fixes and stability work

dot.net/maui

.NET 8 Performance Improvements in .NET MAUI

New features

- `AndroidStripILAfterAOT`
- `AndroidEnableMarshalMethods`
- NativeAOT on iOS

Build & Inner Loop Performance

- Filter Android `ps -A` output with `grep`
- Port WindowsAppSDK usage of `vcmeta.dll` to C#
- Improvements to remote iOS builds on Windows
- Improvements to Android inner-loop
- XAML Compilation no longer uses `LoadInSeparateAppDomain`

Memory Leaks

- Memory Leaks and Quality
- Diagnosing leaks in .NET MAUI
- Patterns that cause leaks: C# events
- Circular references on Apple platforms
- Roslyn analyzer for Apple platforms

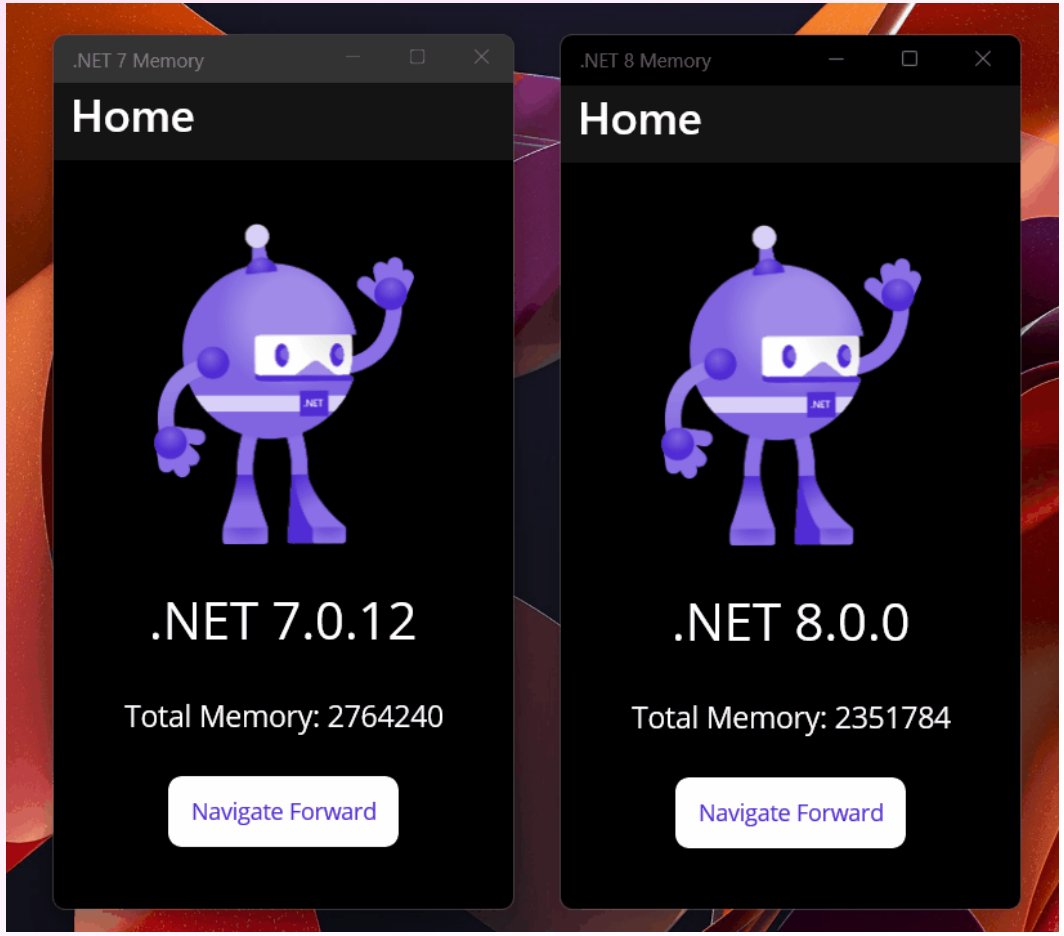
Tooling and Documentation

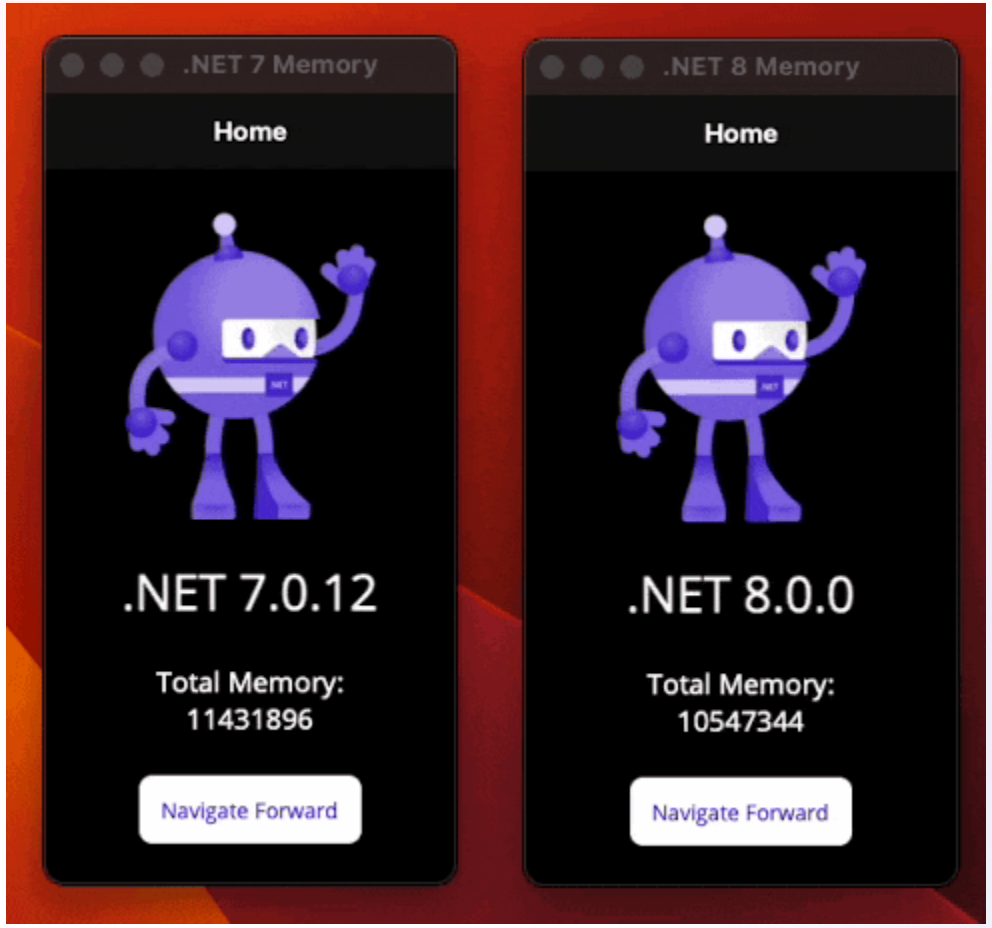
- Simplified `dotnet-trace` and `dotnet-dsrou`
- `dotnet-gcdump` Support for Mobile

Performance or App Size Improvements

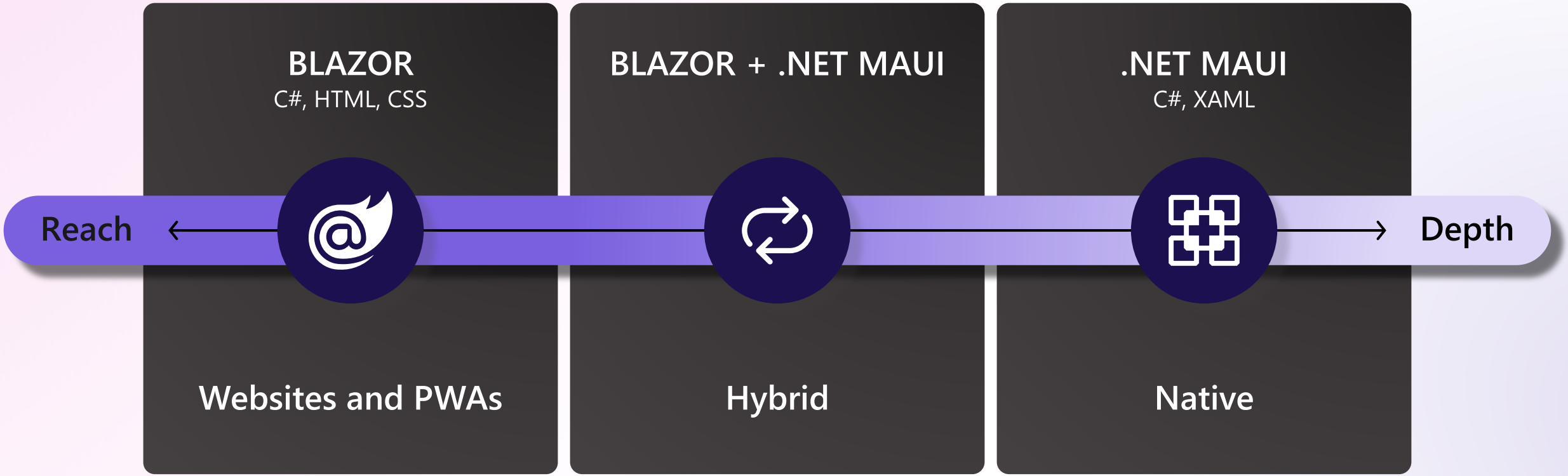
- Structs and `IEquatable` in .NET MAUI
- Fix performance issue in `{AppThemeBinding}`
- Address `CA1307` and `CA1309` for performance
- Address `CA1311` for performance
- Remove unused `ViewAttachedToWindow` event on Android
- Remove unneeded `System.Reflection` for `{Binding}`
- Use `StringComparer.Ordinal` for `Dictionary` and `HashSet`
- Reduce Java interop in `MauiDrawable` on Android
- Improve layout performance of `Label` on Android
- Reduce Java interop calls for controls in .NET MAUI
- Improve performance of `Entry.MaxLength` on Android
- Improve memory usage of `CollectionView` on Windows
- Use `UnmanagedCallersOnlyAttribute` on Apple platforms
- Faster Java interop for strings on Android
- Faster Java interop for C# events on Android
- Use Function Pointers for JNI
- Removed `Xamarin.AndroidX.Legacy.Support.V4`
- Deduplication of generics on iOS and macOS
- Fix `System.Linq.Expressions` implementation on iOS-like platforms
- Set `DynamicCodeSupport=false` for iOS and Catalyst

GC.GetTotalMemory(): .NET 7 vs .NET 8





.NET offers choice for developers



Blazor hybrid apps

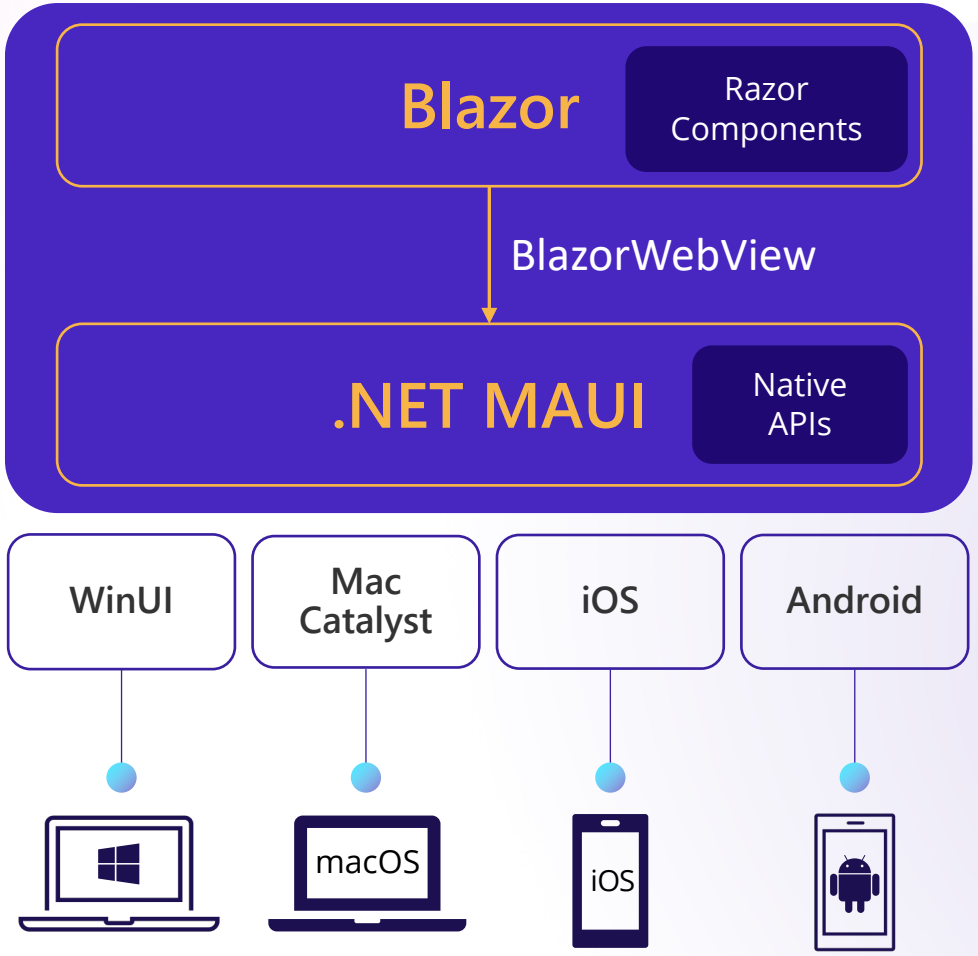
Native app container & embedded controls

Fully native web UI through platform WebView

Multi-platform with MAUI abstractions

No internet required

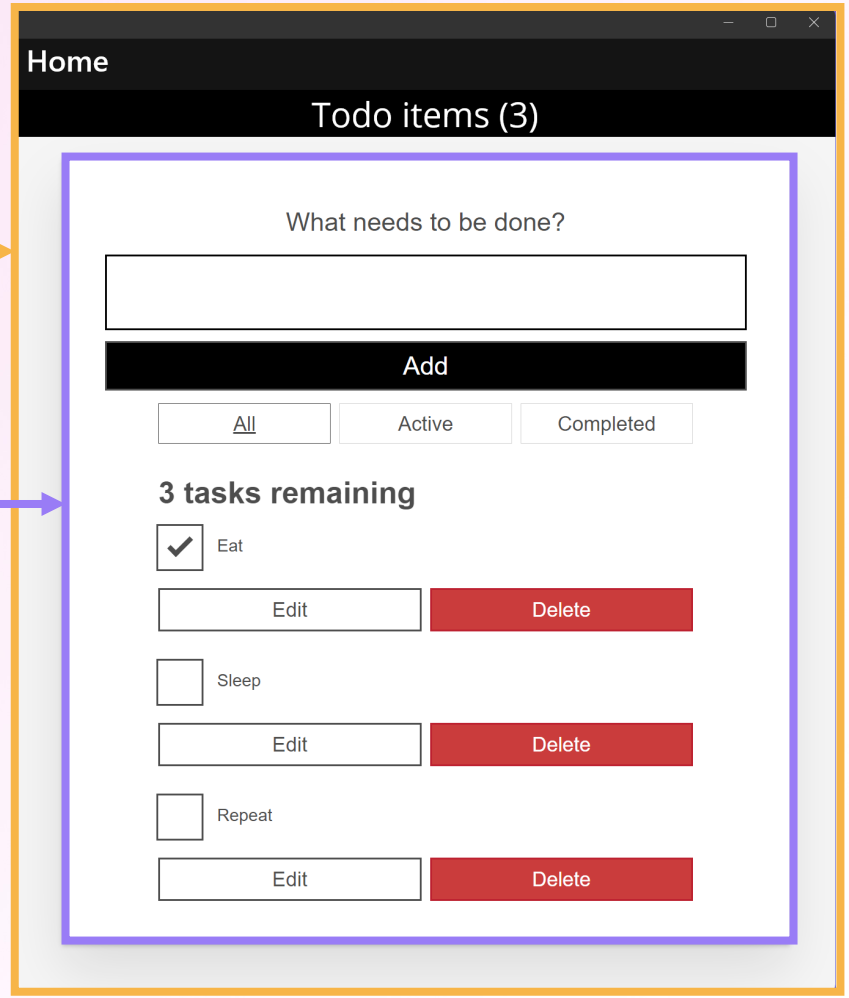
Store distribution and reach



.NET MAUI HybridWebView Experiment

Native app container & controls

Hybrid WebView React JS



- Multi-platform control based on .NET MAUI WebView
- Host static web assets (HTML, JS, CSS, etc.)
- Communicate between JavaScript & .NET
- Reuse your Angular, React, etc. investments in MAUI native desktop and mobile apps
- Use any web-based UI and it can interact with C# code

See more tomorrow @ 3:30 PST!

Desktop Development

.NET 7

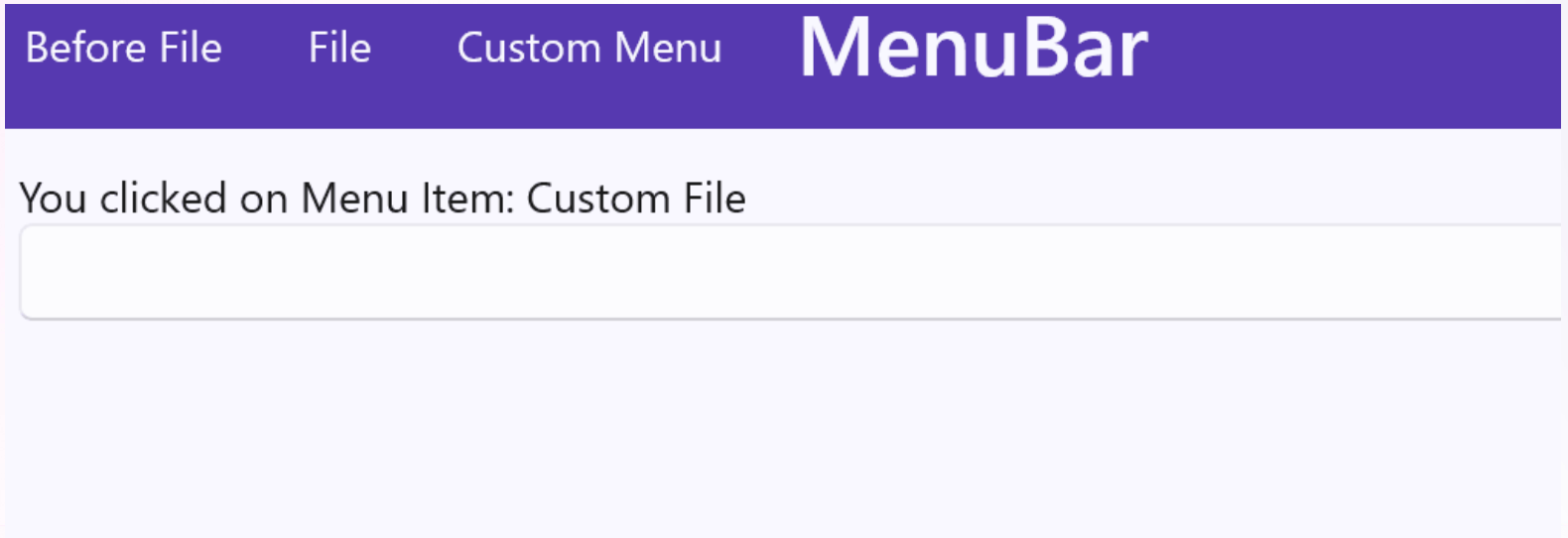
- Tooltips
- Pointer hover
- Right-click & context menus

.NET 8

- Custom keyboard accelerators
- Drag and drop gestures
- Pointer gestures for pressed & released

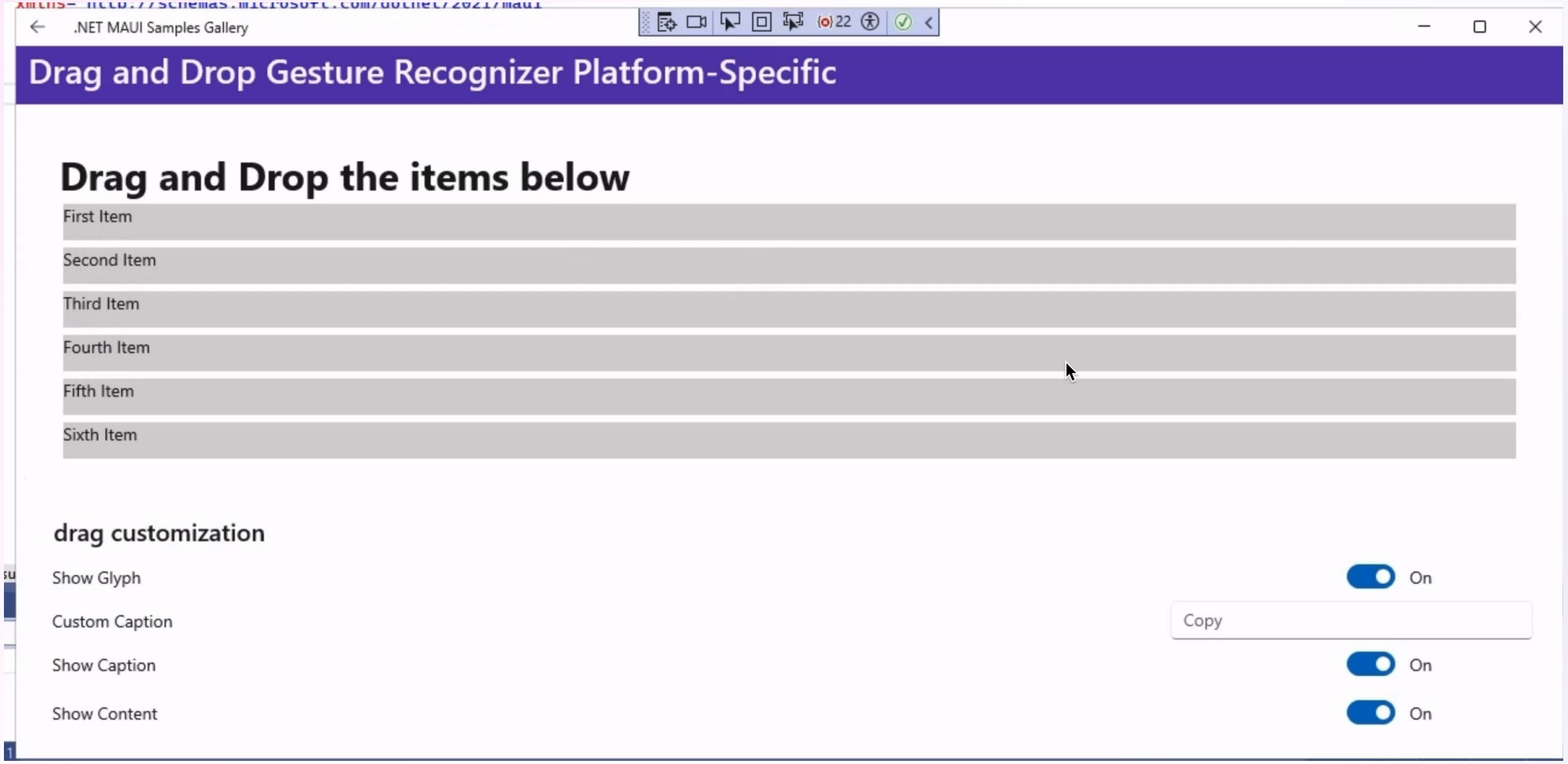
Keyboard Accelerators

```
<MenuFlyoutItem Text=”Custom File">
  <MenuFlyoutItem.KeyboardAccelerators>
    <KeyboardAccelerator Key=”F" Modifiers=”Ctrl">
  </MenuFlyoutItem.KeyboardAccelerators>
</MenuFlyoutItem>
```



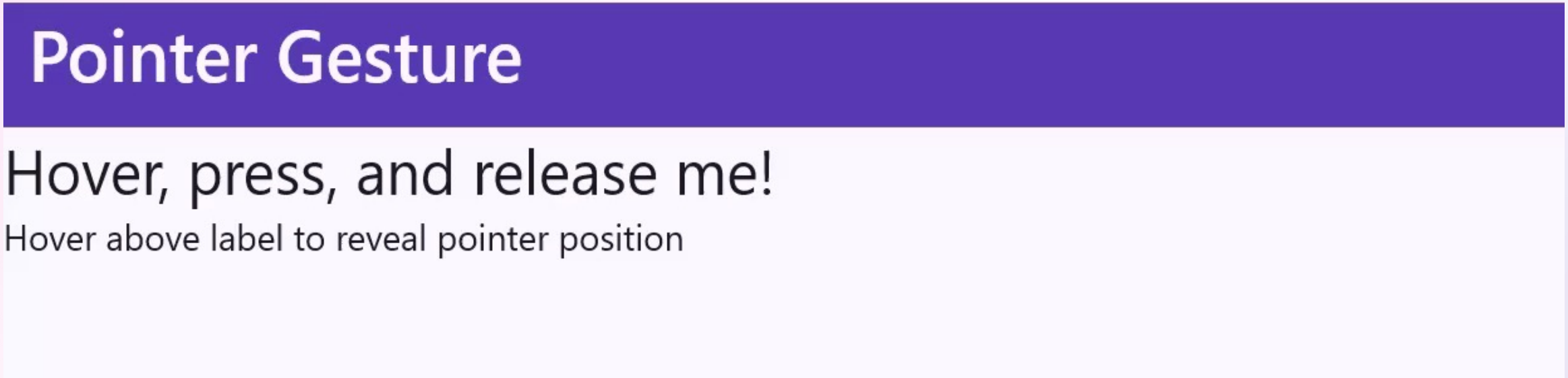
Drag & Drop

Exposed WinUI Drag & Drop APIs such as custom glyphs (images) and captions on drag



Pointer Gestures

PointerPressed and **PointerReleased**



Desktop in .NET 8

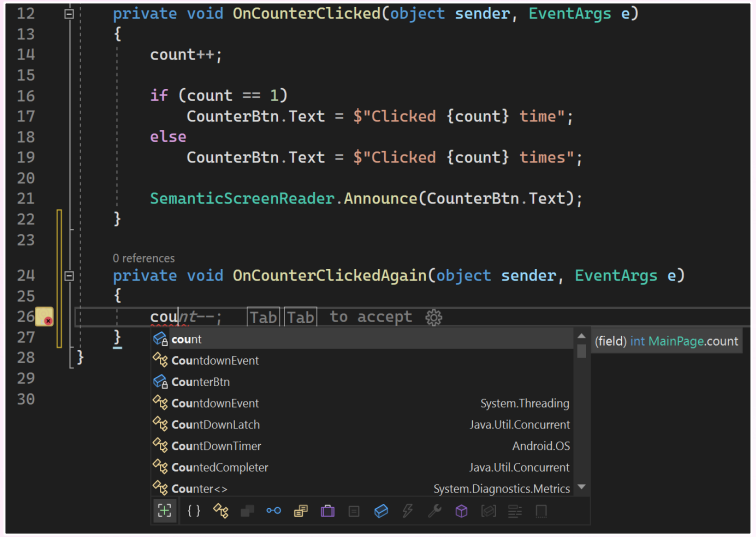
Demo

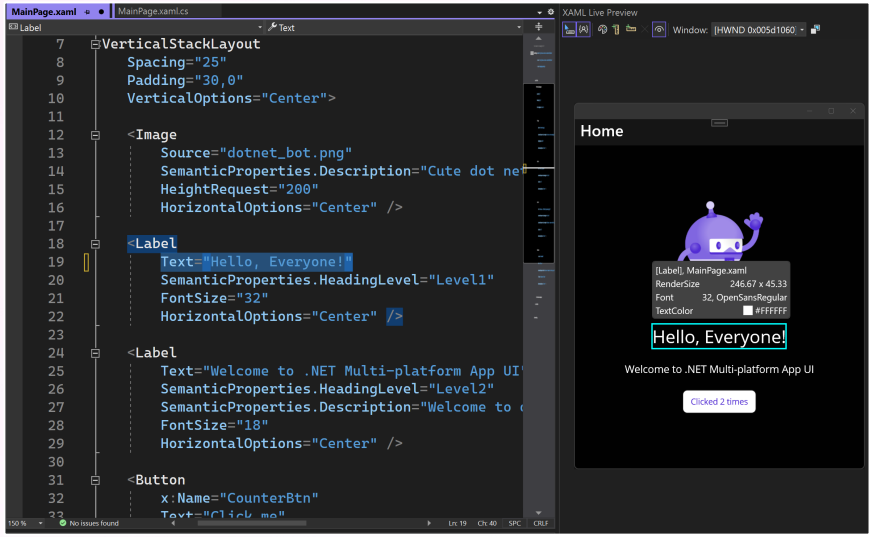


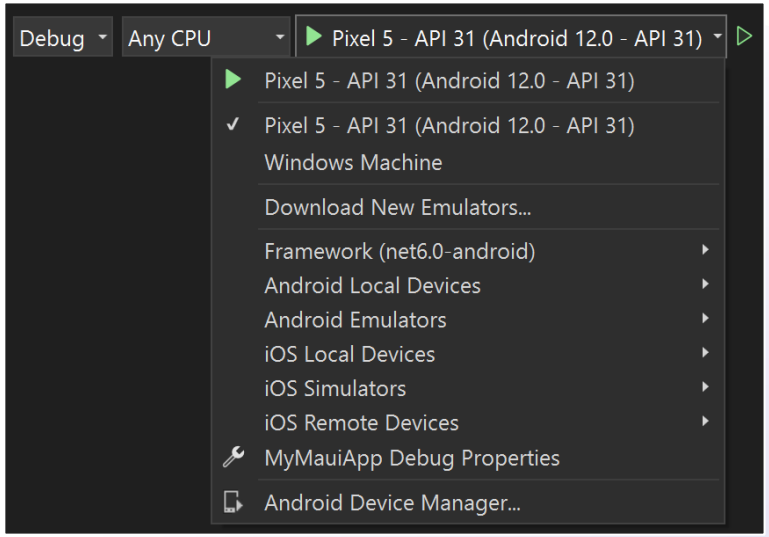


aka.ms/csdevkit
aka.ms/mauidevkit

Visual Studio 2022 productivity







Intelligent code editing

AI-assisted development
Live Share remote collaboration
C#, XAML, Razor IntelliSense

Rapid iteration

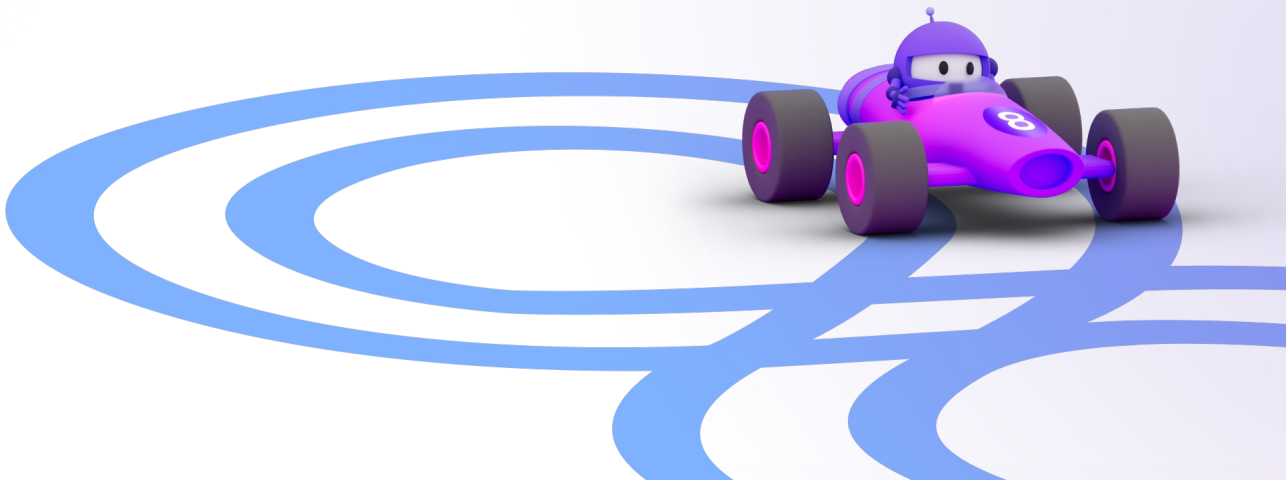
Hot reload XAML, C#, Razor, CSS, and more
Preview and Inspect UI at debug time

Develop for anything, anywhere

Use Android emulators & devices
Develop iOS directly from Windows
Use iOS simulators via Xcode

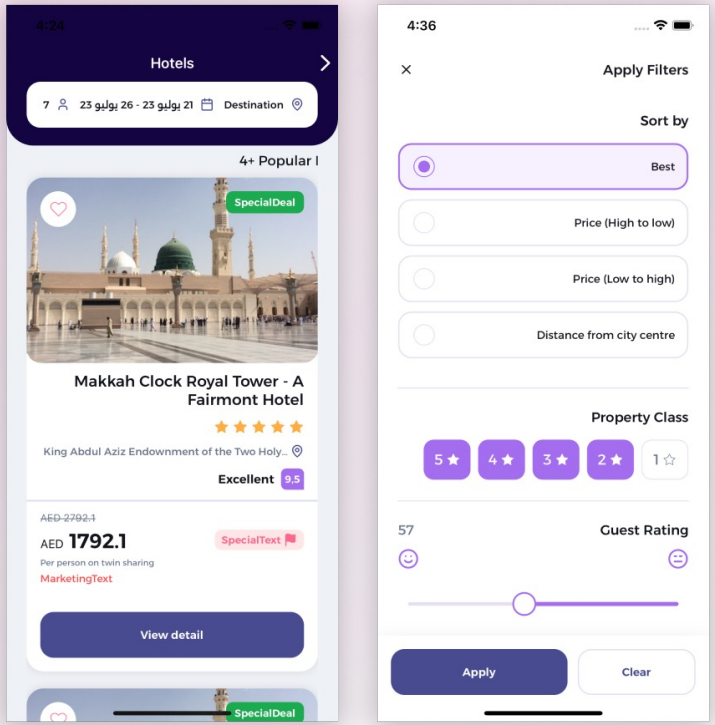
Hot-off-the-press in VS

Demo

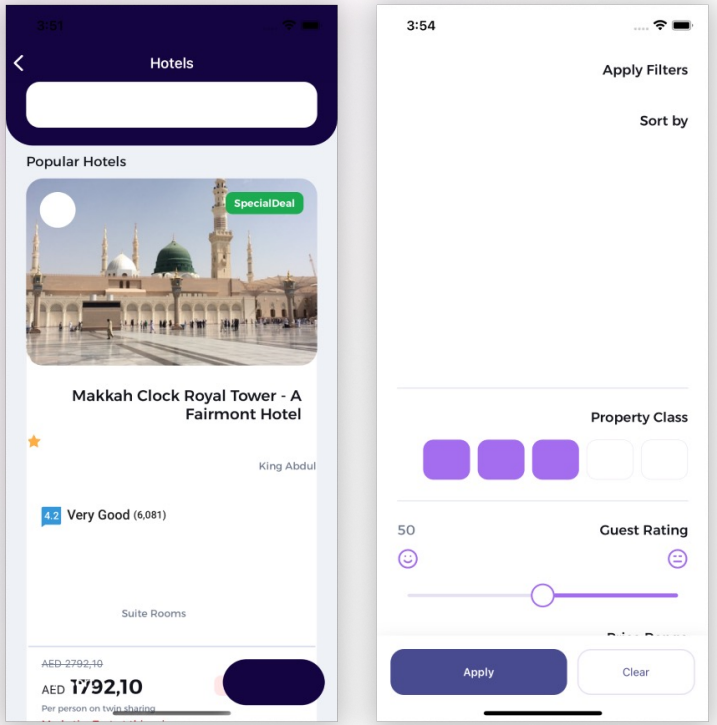


Right-to-left (RTL) Improvements

.NET 8



.NET 7



"Creating Right to Left layouts in .NET MAUI is now a reality thanks to .NET 8. Visual elements are perfectly arranged and Carousels work great now."

– Leo, UX Divers

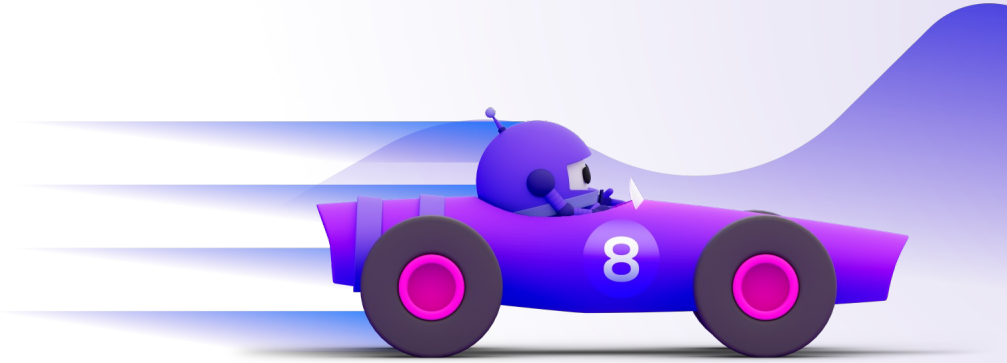
Last but not least

Fixes related to…

- CollectionView
- TabBar visibility
- RoundRectangle Borders
- Android text alignment
- Drag-and-drop functionality
- Tab bar appearance
- CheckBox
- RefreshView
- SwipeItem
- Label
- Button
- Border clipping
- Window glitches
- Image loading problems
- Cursor preservation in text boxes
- Menu key accelerators
- InputTransparent behavior permutations
- Fonts
- Navigation
- Tabs
- File picker
- Entry visibility under keyboard
- iOS SafeArea
- Downscaled iOS images
- Removal of Rosetta
- Unpackaged WinUI apps
- Width & Height requests
- Platform consistency
- NREs & crashes

…and much more!!!

On to .NET 9



Lots of Goals

- Increase delivery velocity of bug fixes
- Reduce build times
- Improve runtime performance
- Support more native desktop experiences
- Improve hot reload
- Swift Interop
- Add more documentation and samples

https://github.com/dotnet/maui/discussions

Start using .NET 8

get.dot.net



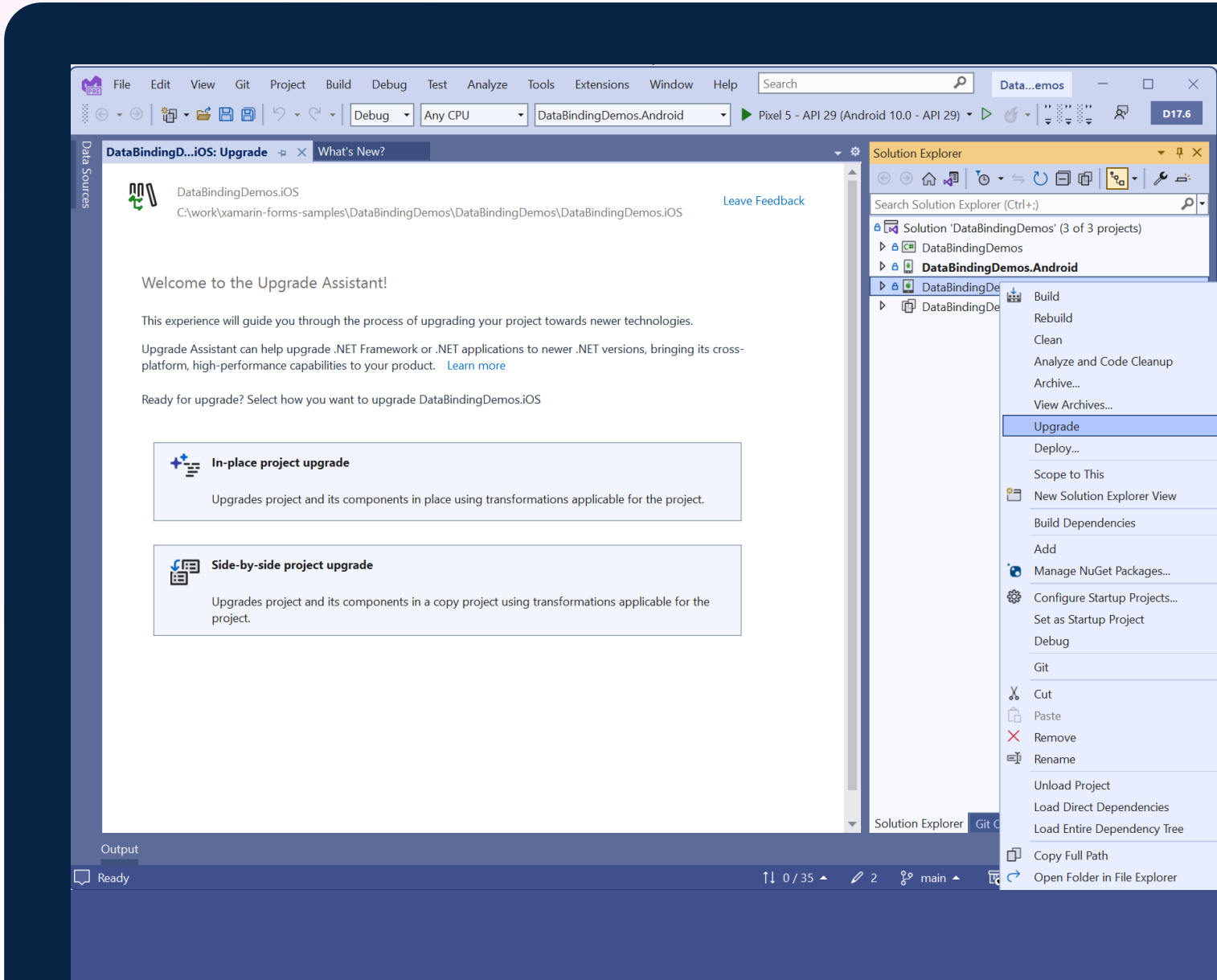
.NET Upgrade Assistant for Xamarin Projects

Highlights:

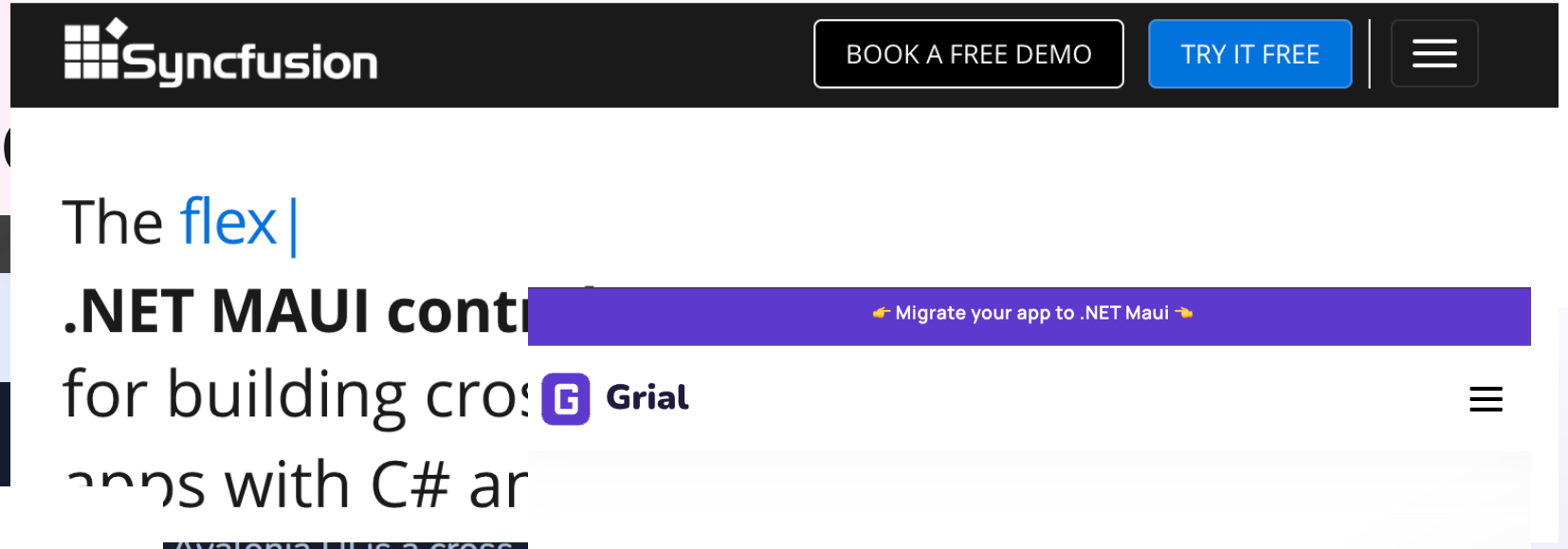
- Converts projects
- Replaces known NuGets
- Fixes namespaces
- Fixes base types
- Applies various code updates

Also can help you upgrade from .NET 6/7 to .NET 8!

maui-upgrades@microsoft.com



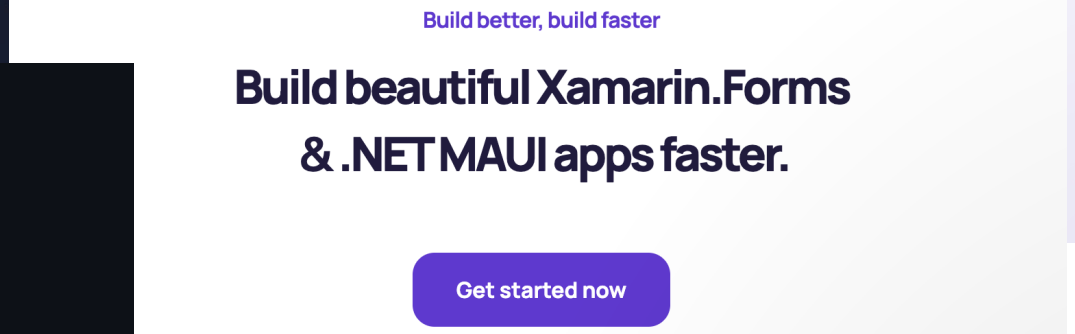

ReactiveUI
Documentation
Api
Contribute
Rea
vs.
ReactiveUI
An advanced, composable, functional reactive model-view-viewmodel framework for all .NET platforms!
Get Started

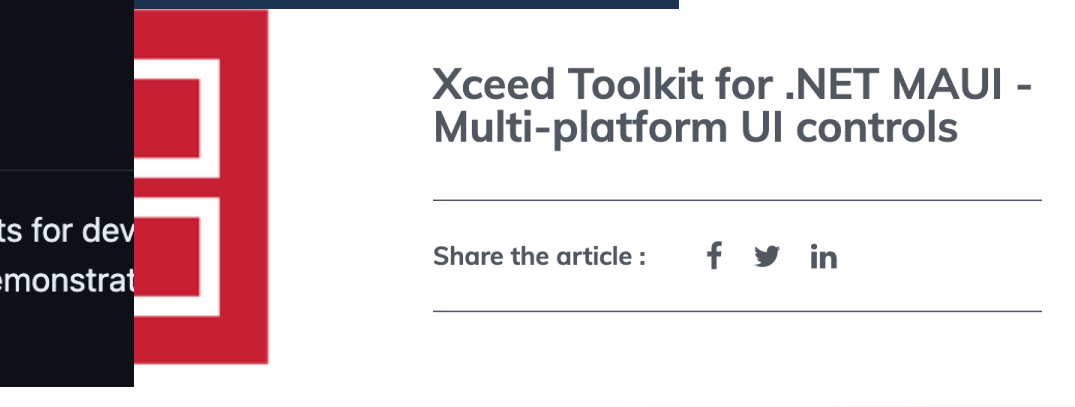

Syncfusion
BOOK A FREE DEMO
TRY IT FREE
The flex|
.NET MAUI

Migrate your app to .NET Maui
Grial


Telerik UI for .NET MAUI
VISUAL STUDIO 2022, .NET
Kickstart your cros
legacy projects wit
components!

.NET foundation
Azure Pipelines
succeeded
nuget
v6.1.0
859.8k
.NET MAUI Community Toolkit
The .NET MAUI Community Toolkit is a collection of common elements for dev
developers tend to replicate across multiple apps. It simplifies and demonstrat
building apps with .NET MAUI.


Build better, build faster
Build beautiful Xamarin.Forms & .NET MAUI apps faster.
Get started now


Xceed Toolkit for .NET MAUI - Multi-platform UI controls
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.NET MAUI at .NET Conf

- Student Zone Intro Workshop
- **Performance Improvements in .NET 8, ASP.NET Core, and .NET MAUI |** David Fowler, Stephen Toub, Jonathan Peppers
- **Build hybrid apps with .NET MAUI |** Beth Massi, Eilon Lipton
- **Spatial Data with Entity Framework Core and .NET MAUI |** Giorgi Dalakishvili
- **Reactive programming with .NET MAUI |** Enis Necipoğlu
- **Design UI agnostic cross-platform applications with .NET |** Dan Ciprean Ardelean
- **Introducing Grial Studio for .NET MAUI |** Diego Rivero, Pablo Germano
- **Building 3D apps with .Net MAUI and Evergine |** Jorge Cantón, Javier Suárez Ruiz
- **Community Toolkit Roundup |** Michael Hawker, Gerald Versluis, Sergio Pedri
- **How to build a story generator application for children with .NET MAUI and Microsoft Azure |** Héctor Uriel Pérez Rojas
- **Build .NET MAUI Apps with DevOps |** Sweekriti Satpathy, David Ortinau
- **Unleashing Cross-Platform Magic in .NET 8: Creating .NET MAUI Apps on Linux with Visual Studio Code |** Bruno Capuano, Luis Beltran

Thank you!

dot.net/maui